

Class 26: Proportions

Determine if the proportions are true.

① $\frac{3}{5} \stackrel{?}{=} \frac{12}{20}$ ② $\frac{10}{22} \stackrel{?}{=} \frac{1}{2}$ ③ $\frac{4}{6} \stackrel{?}{=} \frac{2}{3}$

④ $\frac{3}{5} \stackrel{?}{=} \frac{8}{18}$ - ⑤ $\frac{10}{12} \stackrel{?}{=} \frac{40}{45}$ ⑥ $\frac{20}{35} \stackrel{?}{=} \frac{8}{14}$

⑦ $\frac{12}{16} \stackrel{?}{=} \frac{9}{12}$ ⑧ $\frac{72}{81} \stackrel{?}{=} \frac{21}{27}$ - ⑨ $\frac{8 \text{ miles}}{24 \text{ hours}} \stackrel{?}{=} \frac{7 \text{ miles}}{23 \text{ hours}}$

⑩ $\frac{\$8}{15 \text{ lbs}} \stackrel{?}{=} \frac{\$4}{8 \text{ lbs}}$

⑪ 16 feet for every 8 jumps is equivalent to 20 feet for every 10 jumps?

⑫ $\frac{30}{6} \stackrel{?}{=} \frac{14}{3}$

⑬ $\frac{9}{3} \stackrel{?}{=} \frac{20}{5}$

Solve the Proportions

⑭ $\frac{8}{n} = \frac{24}{27}$

⑮ $\frac{3}{15} = \frac{a}{5}$

⑯ $\frac{42}{63} = \frac{6}{b}$

⑰ $\frac{n}{3} = \frac{12}{18}$

⑱ If a bird flies 22 miles in 55 minutes, how long would it take this bird to fly 2 miles? Write a proportion statement and use it to solve this problem.

⑲ If 40 calories of energy are required to run 90 feet, how many calories would be required to fly 9 feet? Write a proportion statement and use it to solve this problem.

$$- \textcircled{20} \frac{9}{4} = \frac{18}{n}$$

$\textcircled{21}$ If there are 5 ounces of silver for every 2 ounces of gold, how many ounces of gold would be present with 30 ounces of silver? Write a proportion statement and use it to solve this problem.

$$\textcircled{22} \frac{b}{14} = \frac{8}{7}$$

$\textcircled{23}$ If there are 2 gallons of red paint for every 7 gallons of white paint, how many gallons of red paint would be present with 21 gallons of white paint? Write a proportion statement and use it to solve this problem.

$$\textcircled{24} \frac{100}{n} = \frac{10}{9}$$

$$- \textcircled{25} \frac{n}{5} = \frac{7}{8}$$

$$\textcircled{26} \frac{4}{5} = \frac{6}{9}$$

$$\textcircled{27} \frac{7}{2} = \frac{b}{3}$$

$$\textcircled{28} \frac{2}{c} = \frac{5}{9}$$

$\textcircled{29}$ If a submarine travels six miles in 11 hours, how much time would be necessary for the submarine to travel 3 miles? Write a proportion statement and use it to solve this problem.

$\textcircled{30}$ If 20 pounds of meal are required to feed 9 horses, how much meal would be required to feed 4 horses? Write a proportion statement and use it to solve this problem.

$$- \textcircled{31} \frac{3}{7} = \frac{b}{2}$$

$$\textcircled{32} \frac{n}{4} = \frac{1}{9}$$

$$- \textcircled{33} \frac{8}{c} = \frac{11}{7}$$

$$\textcircled{34} \frac{36}{61} = \frac{9}{n}$$

$\textcircled{35}$ The ratio of Jim's coins to Arnold's coins is $\overline{5:3}$. If Jim has 20 coins, how many coins does Arnold have?

③⑥ If the ratio of Rolex watches to Casio watches is 2 to 7, and there are 28 Casio watches, how many Rolex watches are there?

③⑦ The ratio of bluebirds to sparrows is 8:7 and there are 48 bluebirds. How many sparrows are there?

③⑧ If the ratio of the number of cups of flour to the number of cups of oil in a recipe is 4:3, how many cups of oil will be necessary if there are 16 cups of flour?

③⑨ The ratio of hats to jackets is 2:9. How many hats are there if there are 90 jackets?

④① If the ratio of bagels to donuts is 2 to 5, and there are 15 donuts, how many bagels are there?

④① The ratio of Chevy trucks to Ford trucks to Dodge trucks is 3:8:2. If there are 32 Ford trucks, how many Chevy and Dodge trucks are there?

④② If the ratio of dimes to nickels to quarters is 2:1:5, and there are 20 quarters, how many dimes and nickels are there?

④③ The ratio of pencils to erasers to pens is 8:7:2. If there are 24 pencils, how many erasers and pens are there?

④④ There is a container with mixed nuts. If the ratio of almonds to cashews to walnuts is 2:2:3, and there are 10 cashews, how many of the other types of nuts are there?

Challenge Problems

45) If the ratio of the number of paper clips to the total number of paper clips and staples is 4 to 11, and there are 35 staples, how many paper clips are there?

46) The ratio of the number of hamsters to the difference between the number of rats and hamsters is 9:2. How many rats are there if the number of hamsters is 81?

47) If the ratio of the number of bags to the number of cups is 6:7, and there are 60 bags, what is the sum of the number of cups and the number of bags?

48) If the ratio of the difference between the number of red balloons and the number of green balloons to the number of green balloons is 5:8, and there are 24 green balloons, how many red balloons are there?

49) The ratio of the perimeter of a fence and the sum of the perimeters of the same fence and a wall is 5:12, and the perimeter of the fence is 10 ft. What is the perimeter of the wall?

50) If the ratio of Donny's cash to the sum of Donny and Melissa's cash is 4:11, and the sum of their money is \$33, how much cash does Melissa have?

51) If the ratio of the number of bubble gum sticks to the difference between the number of bubble gum sticks and jelly beans is 9:1, what is the total number of bubble gum sticks and jelly beans if there are 40 jelly beans?